

Efficient hydrolysis of glycosides by *Pichia kudriavzevii* with high-activity β -glucosidase: a critical role in enhancing terpene aroma in beer

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ABSTRACT

Beer aroma is strongly influenced by terpenes, many of which exist in hops as non-volatile glycosides. β -Glucosidase (BGL) can release these bound terpenes, but its role in brewing remains underexplored. In this study, twelve non-*Saccharomyces* yeasts were screened, and *Pichia kudriavzevii* KD4 was identified as the best BGL producer. Co-fermentation with *Saccharomyces cerevisiae* US05 under optimized conditions enhanced floral and fruity aromas of beer, increasing esters and terpene alcohols, particularly linalool, nerol, and geraniol. The BGL from KD4 (*PkBGL*) was heterologously expressed in *P. pastoris*, exhibiting optimal activity at 50 °C and pH 5.5, and showing stability under low-temperature, acidic, ethanol-, and iso α -acid-rich conditions. Enzymatic assays confirmed efficient hydrolysis of hop glycosides, while molecular docking revealed stable binding of linalyl and nerol glycosides via hydrogen bonding and hydrophobic interactions. These results highlight *PkBGL* as a promising tool for aroma enhancement in beer through selective terpene release.

1. Introduction

Beer flavor is one of the most important determinants of product quality and consumer acceptance. In addition to its refreshing taste and moderate alcohol content, beer's appeal is largely determined by its complex aroma, shaped by raw materials, yeast metabolism, and brewing processes (Svedlund et al., 2022). Among the diverse chemical groups contributing to aroma, terpenes are particularly important for imparting floral, fruity, and citrus-like notes, especially appreciated in hoppy beer styles (Haslbeck et al., 2018; Schubert et al., 2022). However, most hop-derived terpenes are present as glycosidically bound, non-volatile precursors that do not directly contribute to sensory perception. Hydrolyzing these glycosides through glycosidase has become an attractive approach to enhance beer aroma and diversify flavor profiles without altering hop dosage (Svedlund et al., 2022). In this context, yeast metabolism is of central importance, as yeasts in brewing not only ferment wort sugars into ethanol and carbon dioxide but also contribute to the liberation or biotransformation of aroma-active compounds.

In traditional brewing, *Saccharomyces cerevisiae* is the dominant yeast species due to its high ethanol productivity, robustness, and ability to complete fermentation efficiently. Nevertheless, its ability to enhance aroma complexity, particularly through terpene release, is limited because of relatively weak glycosidase activities (Vrinceanu et al., 2025). In recent years, non-*Saccharomyces* yeasts or lactic acid bacteria have gained attention as alternative or co-fermentation partners, offering unique metabolic properties that can enhance the sensory attributes of beer. Species such as *Wickerhamomyces anomalus*, *Hanseniaspora osmophila*, *Saprochaete suaveolens*, *Candida glabrata*, and *Lactiacaseibacillus paracasei* have been shown to improve flavor complexity through higher ester production, acid modulation, and the release of bound volatile precursors (Han et al., 2023; Herkenhoff et al., 2023; Song et al., 2024; Tan et al., 2021). These non-*Saccharomyces* yeasts provide brewers with tools to tailor beer character and introduce new sensory dimensions. However, their contributions to aroma enhancement are often species- and strain-specific, requiring careful evaluation of their enzymatic repertoires.

Among the glycosidases produced by non-*Saccharomyces* yeasts,

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β -glucosidase (BGL; EC 3.2.1.21) is especially noteworthy, as it hydrolyzes glycosidic bonds at terminal non-reducing glucosides or oligosaccharides, releasing glucose glycone (Zhang et al., 2021). In brewing, BGL can catalyze the release of monoterpene alcohols such as linalool, geraniol, nerol, and α -terpineol, which are key contributors to floral and fruity beer aromas (Han et al., 2023; Zhang et al., 2021). Beyond beer, β -glucosidases are widely studied in winemaking, where they have been implicated in the release of varietal aromas from grape glycosides (Zhao et al., 2024). The activity of BGL is influenced by multiple factors, including pH, temperature, sugar concentration, ethanol concentration, and SO_2 level. Despite its potential, the use of BGL in beer brewing remains limited, largely because of uncertainty regarding enzyme stability under brewing conditions and an incomplete understanding of its substrate specificity. In particular, whether certain BGLs can interact selectively with specific glycosides to enable stable and efficient terpene release has not been clearly demonstrated.

In this study, the fermentation capacity of 12 selected non-*Saccharomyces* yeasts was systematically investigated, and the impact of *Pichia kudriavzevii* on beer aroma through co-fermentation experiments was evaluated. The enzymatic properties of BGL from *P. kudriavzevii* (PkBGL) were characterized. Recombinant PkBGL was heterologously expressed in *P. pastoris*, with its tolerance under different conditions analyzed. Molecular docking was applied to evaluate PkBGL interaction with linalyl and nerol glycosides. This work aims to clarify the role of BGL from non-*Saccharomyces* yeasts in terpene release and highlights the potential of non-*Saccharomyces* yeasts as tools for aroma enhancement in beer.

2. Materials and methods

2.1. Yeast strains, materials, and wort preparation

Twelve selected non-*Saccharomyces* yeasts with proven high BGL activity were separated from diverse natural fermentation systems, namely *P. kudriavzevii* (KD2, KD3, KD4, KD5, KD6, KD7, P10, Y26, and PWRJ11) from suancai or wine, *Wickerhamomyces anomalus* (W3), and *Issatchenkia orientalis* (DF1 and DF2) from fermented grains. Strains were identified and preserved at -80°C in the Tianjin Key Laboratory of Industrial Microbiology, Tianjin University of Science and Technology. Yeasts were activated twice in a yeast extract-peptone-dextrose (YPD) liquid medium (Solarbio, Beijing, China) to obtain pure single-colony cultures before experiments. Growth curves were determined by measuring the optical density (OD) values at 600 nm every 2 h at 30°C and $220\times\text{g}$ for 28 h in 50 mL YPD medium (Fig. S1). Commercial *S. cerevisiae* US05 was obtained from Fermentis, France. *Escherichia coli* DH5 α , *P. pastoris* X33 receptor cell and its expression vector pPICZ α A were preserved in our lab. DNA Polymerase and QuickCut enzyme were purchased from Takara Bio (Dalian, China). Reagents for BGL activity assays, such as *p*-nitrophenol (*p*-NP), *p*-nitrophenyl- β -D-glucopyranoside (*p*-NPG), were obtained from Solarbio (Beijing, China), and Ni-NTA Agarose for protein purification was purchased from QIAGEN GmbH (Hilden, Germany).

Wort was prepared using Pilsner malt (Briess, Wisconsin, USA) and saccharified sequentially at 52°C (30 min), 62°C (1 h), 68°C (30 min), and finalized at 78°C (10 min). The wort was then filtered, and hops were added during a 1 h boil at 0.3 g/L after 10 min and 0.7 g/L after 50 min. The wort was then autoclaved sterilized and cooled.

2.2. Fermentation ability and BGL production capacity of selected non-*Saccharomyces* yeasts

Yeasts were evaluated for ethanol, pH, sugar, and iso α -acids tolerance. The YPD liquid medium was adjusted to pH 2–6 (stepwise, $\Delta 1$), ethanol was supplemented from 3% to 11% (v/v, $\Delta 2\%$), tetrahydro-iso α -acid (Tetrahop Gold®, BartHaas, Nürnberg, Germany) was added to reach 100–500 International Bitterness Unit (IBU, $\Delta 100$ IBU), and wort

concentrations were adjusted to $10\text{--}20^\circ\text{P}$ ($\Delta 2^\circ\text{P}$). OD_{600 nm} for every 24 h was applied to all samples to determine growth ability.

Activated yeast suspensions were cultured in YPD liquid medium for 48 h at 30°C under 220g, centrifuged at 6500g at 4°C for 13 min, and the supernatant was collected as a crude enzyme solution. The BGL activity was determined as described in Section 2.7.1. In this study, *P. kudriavzevii* KD4 exhibited the highest BGL activity and was consequently selected for subsequent experiments.

2.3. Beer preparation via co-fermentation

2.3.1. Optimization of co-fermentation protocol

Four factors were considered in the optimal fermentation protocol, namely KD4: US05 inoculation ratio, total inoculation concentration, timing of US05 addition, and main fermentation duration. KD4 and US05 were inoculated into the wort at ratios of 100:1, 10:1, 1:1, 1:10, and 1:100, respectively, with a total inoculum concentration of 5×10^6 CFU/mL. The total inoculation concentration of both strains ranged from 3×10^6 CFU/mL to 1.1×10^7 CFU/mL (stepwise, $\Delta 2\times 10^6$ CFU/mL). US05 was inoculated 0–60 h ($\Delta 12$ h) after KD4. After sequential inoculation, the wort was placed in a 20°C incubator for main fermentation, with the main fermentation lasting 3–11 d ($\Delta 2$ d). Other parameters were controlled when altering a single factor. Post-fermentation was conducted at 4°C for 7 days following the main fermentation. Single-strain fermentations of KD4 and US05 served as controls. Physiochemical and sensory evaluations of the finished beer were conducted to determine the best protocol.

2.3.2. Physiochemical and aroma analysis

The alcohol content, pH, total acidity content, diacetyl content, residual sugar content, and apparent degree of fermentation in beer were determined according to Chinese National Standard GB/T 4928–2008. Aroma profiles were analyzed by a gas chromatography–mass spectrometry (GC–MS) system (8890-5977C, Agilent Technologies, Santa Clara, CA, USA) according to the method previously described by Tan et al. (2021) and Edgar Herkenhoff et al. (2024). A degassed 5 mL beer sample was mixed with 10 μL of 1000 PPM 2-octanol as an internal standard in 20 mL headspace vials. An HP-INNO WAX capillary column (60 m $\times$ 320 μm $\times$ 0.5 μm , Agilent Technologies) was used for GC analysis, and helium was used as carrier gas at a constant flow rate of 1.5 mL/min. The heating procedure was set as follows, initial temperature of 40°C maintained for 3 min, followed by a ramp at $5^\circ\text{C}/\text{min}$ to 120°C and held for 3 min, then a ramp at $3^\circ\text{C}/\text{min}$ to 230°C and held for 5 min. The inlet temperature was 250°C . MS conditions were as follows: electron impact was applied as an ionization source at 70 eV, with the ion source temperature of 230°C and a solvent delay time of 3 min. Qualitative analysis was performed by comparing with the NIST2020 Standard Reference Database to identify the composition. Quantitative analysis was conducted using 2-octanol as the internal standard. The odor activity value (OAV) is calculated as the ratio of the concentration of an aroma compound in a sample to its odor threshold. Odor thresholds for each substance were obtained from the literature, and concentrations were obtained from quantitative results in this study.

2.3.3. Sensory evaluation

The sensory evaluation panel consisted of 10 healthy nonsmoking panelists who were recruited from Tianjin University of Science and Technology (five males and five females, aged 23–29) with rich experience in this field. During the evaluations at room temperature, 20 mL of beer from each sample was poured into a 50 mL clear plastic cup and randomly coded with a 3-digit number. For the protocol optimization of the co-fermented beer, the sensory evaluation was performed using the hedonic analysis. A 20-point sensory evaluation was used, with scores from 0 to 20 (extreme dislike to extreme like). The sensory attributes evaluated included appearance, foam, aroma, taste, and balance, and the sum of the five scores was the overall score. For sensory evaluation of

sequentially fermented beer, quantitative descriptive analysis (QDA) was used. The same panelists shall describe as many sensory attributes of each sample as possible. After discussion, seven sensory attributes were chosen (appearance, foam, floral, fruity, acidity, bitterness, and sweetness). The evaluation score was a 5-point scale from 1 (not perceivable) to 5 (strongly perceivable). The average score of each attribute computed among the 10 panelists was illustrated in a radar diagram.

2.4. Evaluation of PkBGL hydrolytic capacity

To evaluate the hydrolytic capacity of PkBGL on hop essential oils, the crude BGL enzyme solution from KD4 was freeze-dried to obtain a lyophilized enzyme powder. The control group consisted of 5 mL of hop essential oil without added enzyme powder, while the experimental group comprised 5 mL of hop essential oil supplemented with 2 g of enzyme powder. The mixture was incubated on a shaking incubator at 30 °C and 220g for 36 h. Following the reaction, GC-MS analysis was conducted.

2.5. Sequence and structure analysis of PkBGL

Homology search of PkBGL was performed using BLAST against the GenBank database. The PkBGL sequence was modeled using AlphaFold 3, and phylogenetic trees were constructed using MEGA 11.0 software.

For molecular docking, the PkBGL receptor was prepared using AutodockTools 1.5.6 software. The ligands, linalyl glycoside and nerol glycoside, were built using Chem 3D and processed identically to the receptor using AutodockTools 1.5.6. During the docking process, the coordinates and box size for Autodock Vina molecular docking were determined. The number of docking runs was set to 10, with all other parameters kept at their default values. Semi-flexible docking was performed using Autodock Vina 1.1.2. From the results, the conformation with the lowest binding free energy (affinity) was selected as the final docked conformation. Visual analysis of the binding mode was conducted using PyMOL software.

2.6. Heterologous expression of PkBGL in *P. pastoris*

2.6.1. Construction of recombinant strain

The gene encoding BGL was amplified from KD4 genomic DNA using primers 5'-AGAGAGGCTGAAGCTGAATTCGCACCAAGTTGCAATCAATCATAA-3' and 5'-CCGGCTGGGCCACGTGTAATAAATGATATTTGCA-GAACCAGA-3'. The PCR protocol was as follows: denaturation at 95 °C for 30 s, annealing at 68 °C for 30 s, extension at 72 °C for 60 s, 30 cycles. After digesting the purified PCR products and plasmid vector pPICZαA with Dpn I QuickCut enzyme, the BGL gene fragments were connected to the linearized pPICZαA plasmid vector using DNA ligase. The pPICZαA-BGL plasmid was transfected into *E. coli* DH5α strain. The correct recombinant plasmid for pPICZαA-BGL was extracted from *E. coli* DH5α and then electrotransferred into *P. pastoris* X33 to obtain recombinant *P. pastoris* for BGL expression (de Almeida Parizotto et al., 2021).

2.6.2. Expression and purification of PkBGL

The recombinant *P. pastoris* with pPICZαA-BGL was cultured in YPD liquid medium containing bleomycin at 30 °C and 220g for 16 h, and then transferred a 1% inoculum to buffered glycerol-complex medium with yeast extract (BMGY) and incubated under the same conditions for 12 h. The centrifuged cells were resuspended in buffered methanol-complex medium with yeast extract (BMMY) and incubated under the same conditions for 5 d, and the supernatant was collected by centrifugation at 4 °C and 12,000g for 30 min. The collected supernatant was filtered through a 0.45 μm water-based membrane and was added to a gravity column containing Ni-NTA Agarose. The target PkBGL protein was eluted with ultrapure water containing 60 mM imidazole. The collected protein was added to citrate-Na₂HPO₄ buffer at pH 5.0 and centrifuged for 20 min at 4 °C and 4000g using an ultrafiltration tube to

obtain the purified PkBGL. The concentration of purified PkBGL was determined using the Coomassie Brilliant Blue (G-250) method, and the molecular weight of PkBGL was determined by SDS-PAGE, which matched the predicted 37.4 kDa.

2.7. Enzymatic characterization of PkBGL

2.7.1. PkBGL activity assay

First, 200 μL of crude enzyme solution was incubated with 200 μL of 35 mM *p*-NPG and 600 μL of 100 mM citrate-Na₂HPO₄ buffer (pH 5.0) at 50 °C for 10 min. Reaction was terminated with 1 mL of 1 M NaOH, and the *p*-NP in the mixture was determined by measuring the absorbance at 400 nm. The concentration of *p*-NP was calculated according to the *p*-NP standard curve ($Y = 11.232X - 0.0098$, $R^2 = 0.9993$). One unit of PkBGL activity was defined as the amount of enzyme that liberated 1 nmol *p*-NP per minute under the experimental conditions.

2.7.2. Effects of pH and temperature on PkBGL activity

To investigate the effect of different pH values on PkBGL activity, the incubation temperature was maintained at 50 °C, and the pH of the citrate-Na₂HPO₄ buffer used in the enzymatic reaction system was set between 3.0 and 7.5 (Δ 0.5). To determine the effect of temperature on PkBGL activity, the pH of the citrate-Na₂HPO₄ buffer was 5.0, and the enzymatic reaction was carried out in a water bath at temperatures ranging from 20 to 70 °C (Δ5°C). Thermal stability of PkBGL was assessed at 20 °C and 50 °C water bath treatments, with residual activity measured every 30 min. The pH stability of PkBGL was measured at 4 °C for 24 h in buffers spanning pH 3.0 to 7.5 (Δ 0.5). Residual enzyme activity was expressed relative to untreated enzyme activity (100%).

2.7.3. Effects of ethanol and isoα-acid on PkBGL activity

To determine the effects of ethanol and isoα-acid on PkBGL activity, the enzymatic reactions were performed at 4 °C for 3 h with ethanol (0–20%, v/v) or isoα-acid (adjusted to 0–250 IBU). Relative activity was calculated against controls without ethanol or isoα-acid additions.

2.7.4. Enzyme kinetics

The kinetic parameters of PkBGL were determined using *p*-NPG as a substrate. The reaction mixture containing *p*-NPG from 0.1 to 30 mM in citrate-Na₂HPO₄ buffer (pH 5.0, 100 mM), and the PkBGL enzyme activity was measured under the optimal conditions. The kinetic parameters, the Michaelis-Menten constant (*K_m*), maximum reaction rate (*V_{max}*), and turnover number (*K_{cat}*) of PkBGL toward *p*-NPG were calculated using the Lineweaver-Burk plot and Michaelis-Menten equation.

2.8. Statistical analysis

Data were compiled in Microsoft Excel 2021. One-way analysis of variance (ANOVA) and *t*-test were used to determine the significant differences between samples using IBM SPSS Statistics 21.0, with *p* < 0.05 indicating statistical significance. Origin 2021 and Graphpad Prism 9 were applied for plotting. All experiments were performed in triplicate, and all results are presented as the mean ± standard deviation.

3. Results and discussion

3.1. Fermentation ability and BGL production capacity of selected non-Saccharomyces yeasts

The ethanol, pH, sugar, and isoα-acids tolerances of the twelve selected strains are shown in Fig. 1A–D. Strains KD6, KD7, DF1, and DF2 exhibited superior ethanol tolerance, sustaining growth at 11% (v/v), whereas all strains were inhibited at 13% (v/v). No strains survived at pH 2, while KD2, KD3, KD4, KD5, KD6, and KD7 showed superior growth at pH 3 for 24 h, indicating good acid tolerance. At sugar

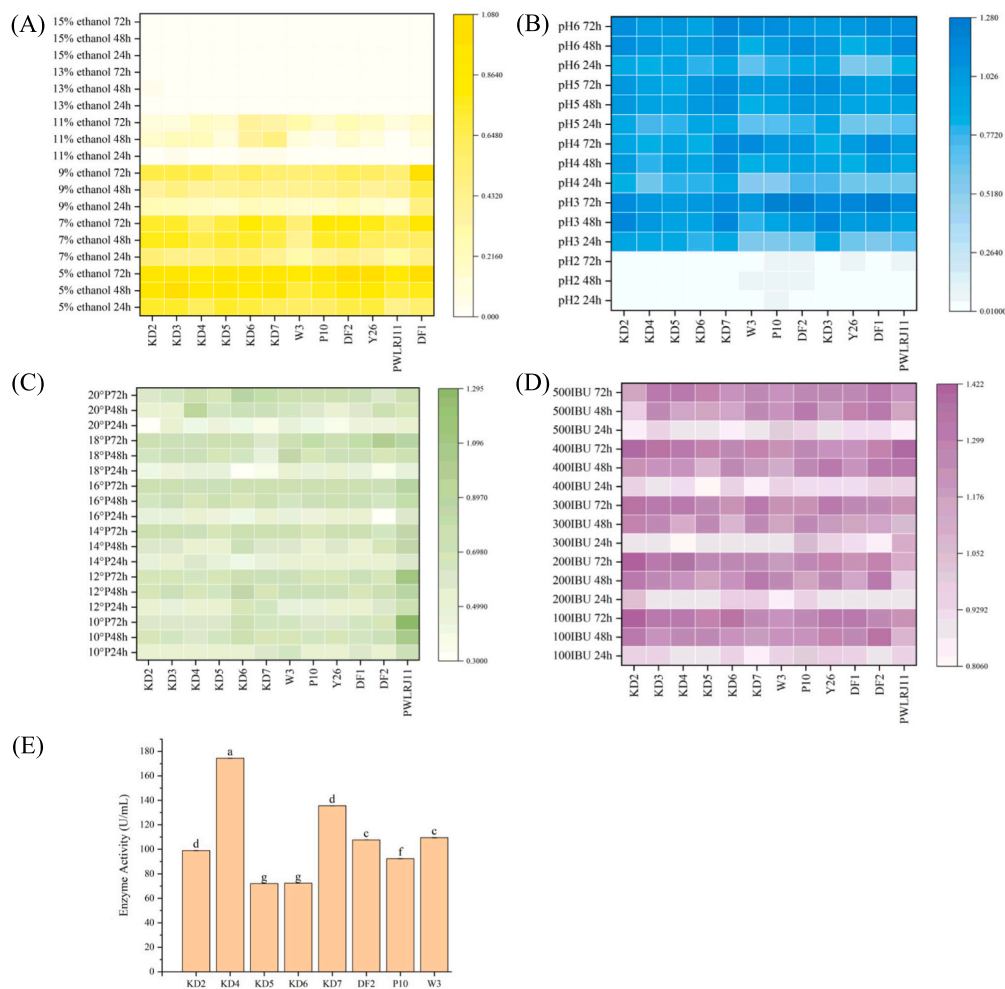


Fig. 1. Heatmap results of yeast fermentation ability selection. (A) The ethanol tolerance for twelve selected strains, with ethanol volume presented in v/v; (B) The pH tolerance for twelve selected strains; (C) The sugar tolerance for twelve selected strains; (D) The iso- α -acids tolerance for twelve selected strains, with iso- α -acids presented in IBU prepared using tetrahydro-iso- α -acid added to YPD medium; (E) The BGL activities for eight selected strains. Different letters indicate significant differences among the groups ($p < 0.05$).

concentrations of 18°C and 20°C, strains KD4, KD6, KD7, and PERJ11 demonstrated better growth capacity, though all strains were suitable for typical beer wort (10°C - 12°C). All 12 non-*Saccharomyces* strains tolerated iso- α -acid levels below 500 IBU, indicating they can withstand the bitterness environment encountered during brewing (about 15–100 IBU). Changes in the aforementioned factors within the yeast growth environment may disrupt the cellular structure of yeast, affecting metabolic activity and reducing yeast viability, leading to an impact on beer fermentation (Lin et al., 2022). Ethanol tolerance is particularly crucial for non-*Saccharomyces* yeasts to function well throughout the fermentation process, even though beer is not traditionally considered a high-alcohol content beverage. Generally, the ethanol content in beer varies from 3% to 5%, however, certain beers such as strong stout or barley wine can reach more than 12% and some Indian pale ale style beers, which feature extremely high hop usage and aroma performance, can reach 10% ethanol content, which makes a high ethanol tolerance yeast more widely beneficial (Schubert et al., 2022).

Growth curves revealed that all strains entered the exponential growth phase at 6–8 h and reached the stationary phase by 18 h (Fig. S1). KD4 achieved the highest cell density, while Y26 exhibited the slowest growth rate. KD3, PWRJ11, Y26, and DF1 entered decline by 20 h, suggesting a shorter stationary phase. Based on growth and stress

tolerance, strains KD2, KD4, KD5, KD6, KD7, W3, P10, and DF2 were selected for further study. The BGL production capacity of the 8 strains showed KD4 had the highest activity (174.39 U/mL), followed by KD7 (135.57 U/mL), whereas KD5 exhibited the lowest (72.11 U/mL) (Fig. 1E). Compared to the BGL production capacity of *Oenococcus oeni* and its acid-tolerant mutant reported by Zhao et al. (2024) (30 U/mL and 70 U/mL, respectively), the non-*Saccharomyces* yeasts in this study exhibited higher enzyme activity. However, when the substrate was changed, the acid-tolerant mutant of *O. oeni* demonstrated a significant increase in BGL production capacity, reaching about 500 U/mL. Overall, KD4, exhibiting the highest BGL activity and optimal growth performance, was selected for co-fermentation experiments.

3.2. Optimization of beer fermentation process

Fig. 2 and Fig. S2 illustrated the effects of inoculation ratio, concentration, US05 inoculation timing, and main fermentation duration on beer sensory and physicochemical characteristics. KD4 single fermentation differed markedly from US05 or co-fermentation, as non-*Saccharomyces* yeast alone lacks sufficient ethanol production capacity to complete fermentation. A higher proportion of *S. cerevisiae* accelerated fermentation and decreased residual sugar but reduced sensory

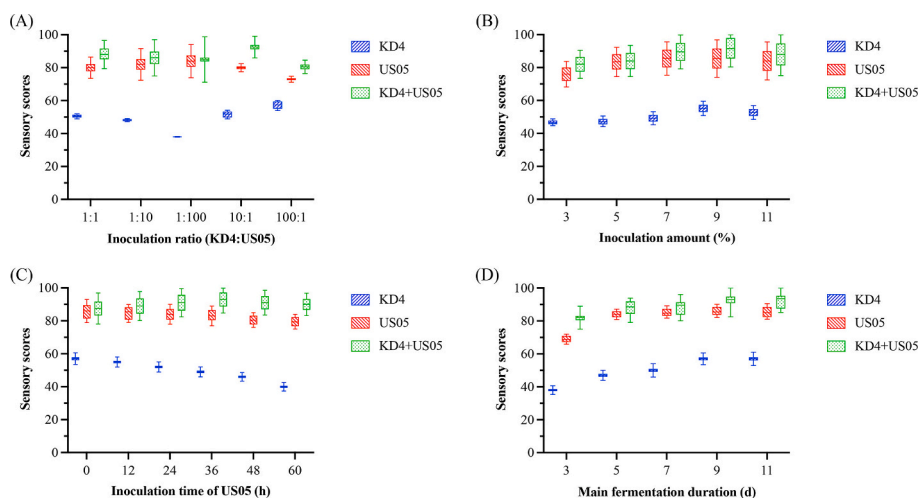


Fig. 2. Optimization of fermentation process. Sensory scores under different inoculation ratios (A), inoculation amount (B), inoculation time of US05 after KD4 (C), and main fermentation time (D).

quality (Fig. S2A). Increasing the proportion of non-*Saccharomyces* yeast contributed to improved sensory quality, with a KD4:US05 ratio of 10:1 achieved the highest sensory score (Fig. 2A). However, when the proportion of KD4 became excessively high, the sensory score of the fermented beer decreased. This decline may be attributed to high concentrations of non-*Saccharomyces* yeast depleting nutrients, thereby inhibiting the growth of US05 and leading to slower fermentation rates and extended fermentation times, which adversely affected beer flavor and taste (Medina et al., 2012). The concomitant lower alcohol content and higher residual sugar levels under this condition corroborated this observation.

Increasing inoculation concentration enhanced fermentation rate, but overly high concentrations increased metabolic by-products, inhibiting new cell proliferation and affecting the sensory quality of beer (Duncan et al., 2023). In this study, an inoculation concentration range of $3\text{--}11 \times 10^6$ CFU/mL was compared, consistent with typical ranges in craft beer production (Callejo et al., 2019; Capece et al., 2018). The highest sensory score was achieved at 9×10^6 CFU/mL (Fig. 2B), correlating with the lowest detected diacetyl content (0.065 mg/L) under this condition (Fig. S2B). Sequential inoculation of US05 significantly influenced the sensory score, alcohol content, pH, total acidity, and diacetyl content (Fig. 2C, Fig. S2C), compared to simultaneous fermentation (US05 inoculation at 0 h). This aligns with findings by Han et al. (2023) in mixed fermentation of *S. cerevisiae* and high-BGL-producing non-*Saccharomyces* yeasts. As the inoculation time of US05 was delayed, the sensory scores of the co-fermented beer gradually increased. The highest sensory score was achieved when US05 was inoculated 36 h after KD4. However, further delay led to a decline in sensory quality, likely due to imbalances in ethanol, sugar, organic acids, and volatile compounds profiles of beer (Li et al., 2022; Su et al., 2024). Extending the main fermentation duration increased sensory scores of co-fermented beers. The diacetyl content in beers fermented with single KD4 and US05, as well as in co-fermentation beers, fluctuated with the main fermentation duration. Diacetyl is a by-product of yeast metabolism, but its level can be decreased through yeast reabsorption during the late main fermentation stage (Spaans et al., 2026). Therefore, an appropriate extension of fermentation time can help to control diacetyl levels. However, excessively prolonged fermentation leads to decreased yeast vitality, hindering diacetyl reduction (Krogerus & Gibson, 2013). The flavor threshold for diacetyl in beer is usually reported as around 0.1–0.4 mg/L (Krogerus & Gibson, 2013), indicating that diacetyl levels in all co-fermented beers in this study remained below perceptible thresholds.

After a comprehensive evaluation, three conditions each from the

inoculation ratio, inoculation concentration, US05 inoculation time, and main fermentation time were selected for the L_9 (3^4) orthogonal test (Table S1). According to the orthogonal test results, the optimal conditions for the co-fermentation process were determined as follows: US05 was inoculated 36 h after KD4, a main fermentation period of 11 d, an inoculation ratio (KD4:US05) of 10:1, and an inoculation concentration of 1.1×10^7 CFU/mL. Fermentation validation under these optimal conditions yielded sensory evaluation and physicochemical indicators for the beer, as shown in Table S2, consistent with the orthogonal test results. Consequently, this protocol was selected for subsequent experiments.

3.3. Sequential fermentation analysis

The aroma profile of the selected sequential fermentation protocol was analyzed using GC–MS. A total of 48 aroma compounds were identified and categorized into 18 esters, 8 alcohols, 9 acids, 7 terpenes, 4 aldehydes/ketones, and 2 phenols (Fig. S3). The total volatile compound contents under co-fermentation were similar to those under US05 single fermentation. However, as demonstrated by the PCA results shown in Fig. 3A, the detailed profiles of each aroma compound and the overall aroma characteristics were clearly distinguishable.

Esters are the most abundant aroma compounds in beer. The highest concentrations of ethyl acetate, ethyl octanoate, and phenylethyl acetate were found in co-fermented beers, at 8297 $\mu\text{g/L}$, 1387 $\mu\text{g/L}$, and 3278 $\mu\text{g/L}$, respectively. Compared to US05 single-fermented beer, the levels of these three esters increased by 5.6-fold, 2.4-fold, and 2.1-fold, respectively (Fig. 3B). Generally, yeast can synthesize ester compounds through the catabolic breakdown of branched-chain amino acids followed by enzymatic esterification of the corresponding short-chain fatty acids (Rumpl et al., 2025). This implies that yeast's amino acid metabolic preferences can significantly influence the production of ester compounds (Zhu et al., 2024). The concentration differences in esters observed in this study can be explained by variations in the activity of the enzyme involved in the metabolism of amino acids into esters between *S. cerevisiae* and *P. kudriavzevii* (Li et al., 2020). Additionally, non-*Saccharomyces* yeasts may also stimulate the upregulation of genes encoding certain ester-synthesizing enzymes in *S. cerevisiae* (Zhang et al., 2024). Isoamyl alcohol, isobutanol, and phenethyl alcohol are the main volatile higher alcohol compounds. The concentrations of isobutanol, isoamyl alcohol, and phenethyl alcohol in the co-fermented beer were 135.05 $\mu\text{g/L}$, 624.56 $\mu\text{g/L}$, and 892.18 $\mu\text{g/L}$, respectively. Higher alcohols are by-products of yeast metabolism during beer fermentation. Moderate levels of higher alcohols enhance the richness

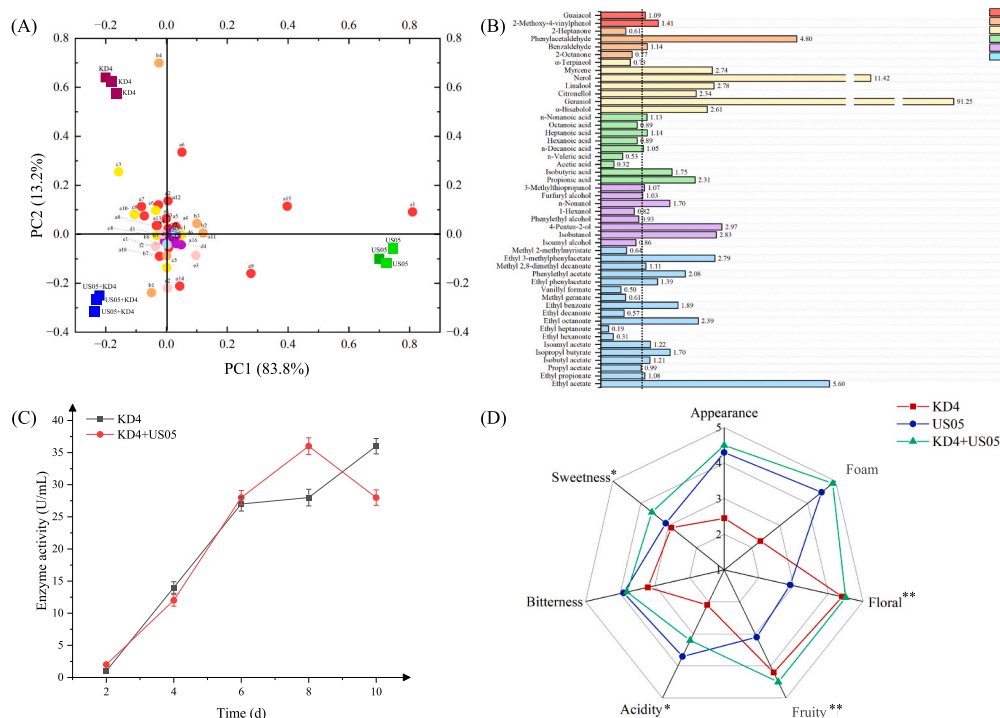


Fig. 3. (A) PCA biplot of main aroma compounds in beers obtained via the single fermentation and co-fermentation of KD4 and US05. The numbered dots represent aroma compounds, in which a1 is ethyl acetate, a2 is ethyl propionate, a3 is propyl acetate, a4 is isobutyl acetate, a5 is isopropyl butyrate, a6 is isoamyl acetate, a7 is ethyl hexanoate, a8 is ethyl heptanoate, a9 is ethyl octanoate, a10 is ethyl decanoate, a11 is ethyl benzoate, a12 is methyl geranate, a13 is vanillyl formate, a14 is ethyl phenylacetate, a15 is phenylethyl acetate, a16 is methyl 2,8-dimethyl decanoate, a17 is ethyl 3-methylphenylacetate, a18 is methyl 2-methylmyristate, b1 is isobutanol, b2 is isoamyl alcohol, b3 is 1-hexanol, b4 is n-nonanol, b5 is furfuryl alcohol, b6 is 4-penten-2-ol, b7 is 3-methylthio propanol, b8 is phenylethyl alcohol, c1 is acetic acid, c2 is isobutyric acid, c3 is n-valeric acid, c4 is propionic acid, c5 is hexanoic acid, c6 is heptanoic acid, c7 is octanoic acid, c8 is n-nonanoic acid, c9 is n-decanoic acid, d1 is myrcene, d2 is linalool, d3 is α -terpineol, d4 is nerol, d5 is citronellol, d6 is geraniol, d7 is α -bisabolol, e1 is 2-octanone, e2 is benzaldehyde, e3 is phenylacetaldehyde, e4 is 2-heptanone, f1 is guaiacol, and f3 is 2-methoxy-4-vinylphenol. (B) Changes in contents of aroma compounds via co-fermentation, with the values in the chart reflecting the ratio of increase/decrease to those via single US05 fermentation. Compounds were categorized in different colors. (C) The BGL activities during the fermentation process of single KD4 fermentation and co-fermentation. (D) Radar map of scores applied via single or co-fermented beers. “*” indicates a significant difference ($p < 0.05$) in the intensity of sensory attributes between single US05 fermentation and co-fermentation; “***” indicates a highly significant difference ($p < 0.01$) in the intensity of sensory attributes between single US05 fermentation and co-fermentation.

and overall flavor complexity of beer (Pires et al., 2014). Compared to beers fermented with single US05, co-fermentation significantly reduced the contents of organic acids, particularly vinegar-like acetic acid and putrid-like valeric acid, indicating that co-fermentation balances the aroma profile of beer.

Compared to single US05 fermentation, the total content of terpenes in co-fermented beers was significantly higher ($p < 0.05$). Specifically, myrcene content increased 1.7-fold, linalool content increased 1.8-fold, nerol content increased 10.4-fold, citronellol content increased 1.3-fold, and geraniol content increased 92.6-fold in co-fermented beer compared to US05. Although terpenes constitute a small proportion of beer's total compounds, their low detection thresholds make them crucial contributors to beer flavor. Further screening of key volatile compounds through OAV calculations identified myrcene (citrus and balsam note), linalool (citrus and lavender note), and geraniol (sweet, floral, and fruity note) as key terpene compounds in co-fermented beer (Table S3). The OAVs of these key terpene compounds in co-fermented beer were significantly higher than those in US05 single-fermented beer, indicating that co-fermentation enhances beer aroma complexity by imparting floral and fruity notes. Almost no significant differences ($p > 0.05$) in aldehydes, ketones, and phenols were observed between single fermented and co-fermented beers, indicating these compounds have a minimal impact on aroma characteristics.

The results of BGL activity measurements during the fermentation process of KD4-fermented beer and co-fermented beer are shown in Fig. 3C. In the KD4 single fermentation, BGL activity exhibited an upward trend with increasing fermentation time and peaked at 36.21 U/

mL on day 10. In the co-fermentation, BGL activity first increased and then decreased, peaking at 35.87 U/mL on day 8. During the early fermentation stage, BGL activity in the KD4-fermented beer was higher than that in the co-fermented beer. This may be attributed to the stable environment of the single-strain fermentation system, allowing KD4 to better adapt and maximize its metabolic potential. These results confirmed that BGL performs a hydrolytic function during fermentation. Specifically, it acted on glycosides in beer to promote their hydrolysis reaction, which increased the content of free aroma compounds in beer and imparted richer floral and fruity aroma characteristics.

The sensory QDA results for beers fermented with single and mixed yeast strains are shown in Fig. 3D. Beers fermented with single KD4 exhibited high intensity of floral and fruity aromas, but low intensity of acidity, appearance, and foam. Compared to KD4 single fermentation, beers fermented with single US05 showed lower floral and fruity aroma intensity, but higher intensity of foam, appearance, acidity, and bitterness. Compared to US05 single fermentation, the co-fermentation exhibited highly significantly higher floral and fruity intensity ($p < 0.01$), and significantly higher sweetness intensity ($p < 0.05$), along with significantly lower acidity intensity ($p < 0.05$). This indicated that co-fermentation enhanced floral and fruity aromas while mitigating acidity, resulting in a more harmonious taste of beer that typically signals higher consumer acceptance (Praia et al., 2022).

3.4. Sequence analysis of *PkBGL*

Homology analysis showed that most of the annotated BGLs with

high sequence similarity to *PkBGL* were derived from *Pichia* yeast (Fig. S4A). The sequence identity of *PkBGL* with the BGL from *P. kudriavzevii* (OUT20650.1) and *P. kudriavzevii* (ONH71489.1) was as high as 98%. The alignment result of the *PkBGL* sequence (GenBank: XP_029319821.1) with the sequence from the GH1 family with resolved crystal structure is shown in Fig. S4B. Conserved regions (marked in red) exhibit high similarity across all sequences and likely represent functional cores, such as catalytic sites or substrate-binding sites. Among these, amino acid residues Asn62, Phe122, Gly157, Gly194, Gly195, Tyr238, Pro239, Leu250, Gly294, Gly309, and Gly315 are highly conserved. The conserved asparagine and phenylalanine residues in BGL

derived from *Trichoderma harzianum* have been reported to be associated with hydrolase and glycosyltransferase functions of BGL, respectively (Florindo et al., 2018). Conserved tyrosine and leucine residues in the BGL from *Humicola insolens*, acting as hydrogen bond acceptors, can interact with BGL ligands to facilitate their release, thereby enhancing product tolerance of BGL (Correa Costa et al., 2019). Furthermore, highly conserved aspartate and glutamate residues in GH1 family BGLs were predicted to be the catalytic nucleophilic and acid/base residues, respectively (Dong et al., 2021; Santos et al., 2016), and were considered crucial active sites facilitating glycoside hydrolysis by GH1 family BGLs (Fusco et al., 2018).

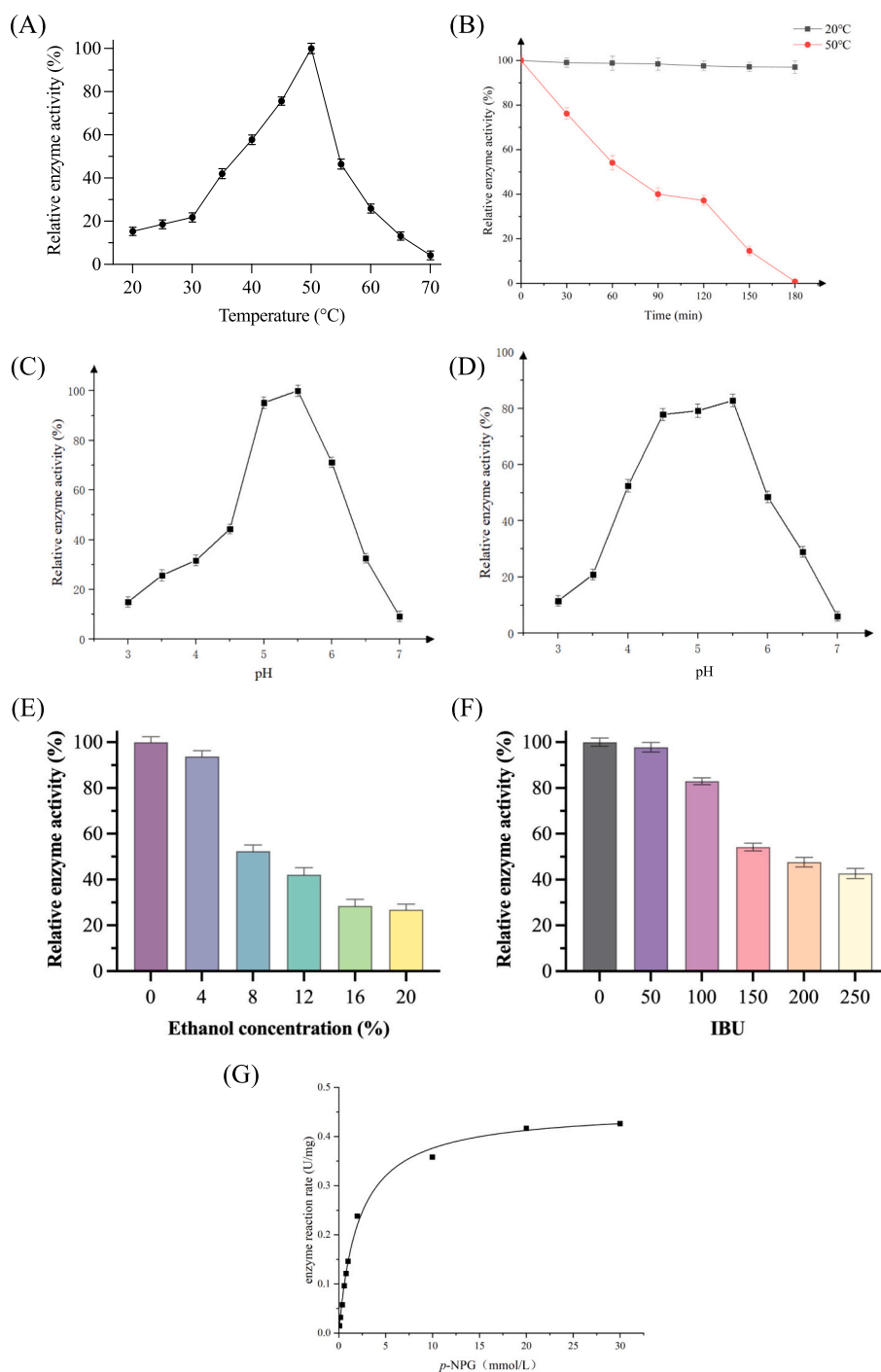


Fig. 4. Effect of temperature (A and B), pH (C and D), ethanol concentration (E), iso α -acid concentration (F) on the activity of the *PkBGL*, and Michells-Menten plotting for *PkBGL*(G).

3.5. Enzymatic characteristics of *PkBGL*

3.5.1. *PkBGL* activities at different conditions of pH and temperature

The gene encoding *PkBGL* in *P. kudriavzevii* KD4 was cloned, the recombinant plasmid pPICZαA-*pkBGL* was successfully constructed and expressed in *P. pastoris* X33, yielding a *PkBGL* protein with a molecular weight of about 40 kDa. Enzyme characteristics directly reflected the adaptability of the enzyme to various environmental conditions and its application potential. pH and temperature are the two primary factors that could impact BGL activity (Kaushal et al., 2021). As shown in Fig. 4A, the optimal temperature of *PkBGL* was 50 °C. When the temperature exceeded 50 °C, the relative activity of *PkBGL* sharply decreased. Moreover, *PkBGL* exhibited better performance at low temperature than at high temperature. In beer brewing, it is preferred to perform main fermentation at low temperature (~20 °C), as low temperature not only facilitates the accumulation and preservation of fruit and fermentation aroma of beer, but also contributes to the ethanol tolerance of non-*Saccharomyces* yeast (Gao & Fleet, 1988). Therefore, the thermal stability of *PkBGL* was further determined at 20 °C and 50 °C (the optimal temperature) (Fig. 4B). Results showed that residual enzyme activity remained above 95% after 3 h reaction at 20 °C, indicating excellent stability. In contrast, residual activity at 50 °C decreased significantly with prolonged reaction time, dropping by half after 1 h and becoming undetectable after 3 h. This demonstrates that *PkBGL* exhibits greater stability under low-temperature conditions and showcases its potential for application in the low-temperature main fermentation stage.

For pH, the optimal value of *PkBGL* was 5.5 (Fig. 4C). Wort is typically fermented in an acidic environment, with a pH of about 5.5 and 4.5 before and after beer fermentation (Canonico et al., 2021). Within this pH range, *PkBGL* maintains relative activity above 45.36%. To further investigate the pH stability of *PkBGL*, its residual enzyme activity was assessed after 24 h of reaction at different pH levels (Fig. 4D). At pH values ranging from 4.0 to 6.0, *PkBGL* maintained residual enzyme activity above 50%, with activity remaining around 80% particularly at pH 4.5 to 5.5. These results indicate the potential application of *PkBGL* in the low-pH fermentation environment of beer fermentation.

3.5.2. Effect of ethanol and isoα-acids on BGL activity

For the application of BGL in beer brewing, the ethanol concentration is another important factor that should be considered. As shown in Fig. 4E, the *PkBGL* activity exhibited a decreasing trend with increasing ethanol concentration. At 4% (v/v) ethanol content, the relative enzyme activity remained above 90%; at 8% (v/v) ethanol content, it still maintained around 50%; at 16% (v/v) ethanol content, it dropped to approximately 30%. Some studies indicate that ethanol at certain concentrations enhances BGL activity. For instance, ethanol concentrations below 10% (v/v) increased BGL activity from *O. oeni* (Zhao et al., 2024); concentrations below 15% (v/v) boosted BGL activity from *Dictyoglomus turgidum* (Fusco et al., 2018); while 1% (v/v) ethanol concentration increases BGL activity from *Metschnikowia pulcherrima* (Mateo, 2023). The enhancement of enzyme activity could be attributed to additional hydrogen bond interactions potentially aiding enzyme stability, though further structural studies are required for confirmation (Fang et al., 2016; Mateo, 2023). However, most native BGL activity was inhibited in the presence of alcohol due to reduced flexibility, diffusion limitations, and destabilization of the enzyme (Fang et al., 2016). The ethanol concentration of finished beer is usually below 10% (v/v), thus demonstrating the potential application of *PkBGL* in beer fermentation.

As the content of isoα-acids increased, the activity of *PkBGL* exhibited a decreasing trend (Fig. 4F). When isoα-acid content ranged from 0 to 100 IBU, the relative activity of *PkBGL* remained above 80%. The relative activity of *PkBGL* could be maintained more than 50% when the isoα-acid content was 150 IBU. As previously mentioned, the content of isoα-acids during beer brewing ranges from 15 IBU to 100 IBU. It has been reported that the fermentation process reduces the content of isoα-

acids through a series of isomerization reactions, influenced by various brewing techniques and brewing microorganisms (Lu, 2015; Sharp et al., 2017; Yang et al., 2021). Therefore, *PkBGL* retained over 50% relative activity at 150 IBU, demonstrating its potential for application in beer brewing. To our knowledge, no studies have reported on the isoα-acid tolerance of BGL.

3.5.3. Kinetics parameters of *PkBGL*

Fig. 4G shows the Michaelis-Menten plot for *PkBGL*, from which the kinetic parameters were derived. Under the optimal conditions, *PkBGL* had a K_m value of 2.14 mM, a V_{max} value of 0.46 U/mg, and a K_{cat}/K_m value of 0.13 mM/s when *p*-NPG was used as the substrate. The K_m value of *PkBGL* corresponds well with the results of studies by Lu et al. (2013) and Saha and Bothast (1996), who determined the K_m values of 2.09 mM and 2.3 mM for BGL using *p*-NPG as the substrate at 50 °C. However, Li et al. (2018) found a K_m value of 0.4 mM with *p*-NPG as substrate at 70 °C, and a K_{cat}/K_m value of 72 mM/s. The discrepancy with the kinetic parameters in this study could be due to the different reaction conditions (e.g., buffer, pH, temperature) and the source of BGL.

3.6. Hydrolytic capacity of *PkBGL* to glycosides

The capacity and characteristics of *PkBGL* in releasing terpene compounds from glycosides of hop were comparatively evaluated by adding *PkBGL* into hop essential oil. As shown in Fig. 5A, three glycosidically bound terpenes were detected in the commercial hop essential oil, namely linalool (lavender-like note), α-terpineol (lilac-like note), and nerol (rose-like note). Compared to untreated samples, the content of linalool and nerol in *PkBGL*-treated samples increased extremely highly significantly ($p < 0.001$) by 2.67-fold and 3.06-fold, respectively, indicating that *PkBGL* exhibited pronounced hydrolytic activity toward these glycosidically bound terpene compounds. As described in Section 3.3, linalool has also been identified as a key aroma compound in co-fermented beer. It is noteworthy that while the sensory impact of nerol on the co-fermented beer was not the most dominant in this study, its concentration in co-fermented beer was significantly higher than in single-fermented beer. Moreover, its contribution could become perceptible in beer styles with higher hop dosages, where a greater abundance of glycosidic precursors allows for a more significant release above its flavor threshold. Previous studies had demonstrated that BGLs from different sources showed different specificity for hop glycosides (Sharp et al., 2017). Most relevant studies have reported that adding the BGL enzyme or inoculating yeast strains with high BGL activity significantly increased the levels of linalool and nerol provided by hops (Cibaka et al., 2017; Passaghe et al., 2024; Sharp et al., 2017), consistent with our findings. In the hop, monoterpene alcohols exist in a glycosidically bonded form. The BGL activity causes the release of an aromatic terpene (and of a glucose molecule) from a nonaromatic terpenyl glycoside (Liang et al., 2022). As terpene alcohols, linalool, α-terpineol, and nerol can all be released from their glycoside forms by BGL and can also be interconverted through reductions, translocations, isomerizations, and cyclizations (Buiatti et al., 2023; Gueguen et al., 1999). In addition, glycosides with primary alcohol aglycones (nerol glycoside) showed faster enzymatic hydrolysis compared to glycosides with tertiary alcohol aglycones (linalool glycoside), which may account for the higher release of nerol compared to linalool (Caffrey et al., 2021).

3.7. Molecular docking and structure insights

The characteristics and functions of the BGLs, such as catalytic rate and substrate specificity, are closely related to their basic structures (Yang et al., 2025). AlphaFold 3 yielded a high-confidence (99.9%) 3D structure of *PkBGL* (Fig. 5B). To further analyze the binding mode and interactions between the substrate and *PkBGL*, molecular docking was performed using the previously described glycosidically bound linalool and nerol as ligands against the predicted 3D structure of *PkBGL*. The

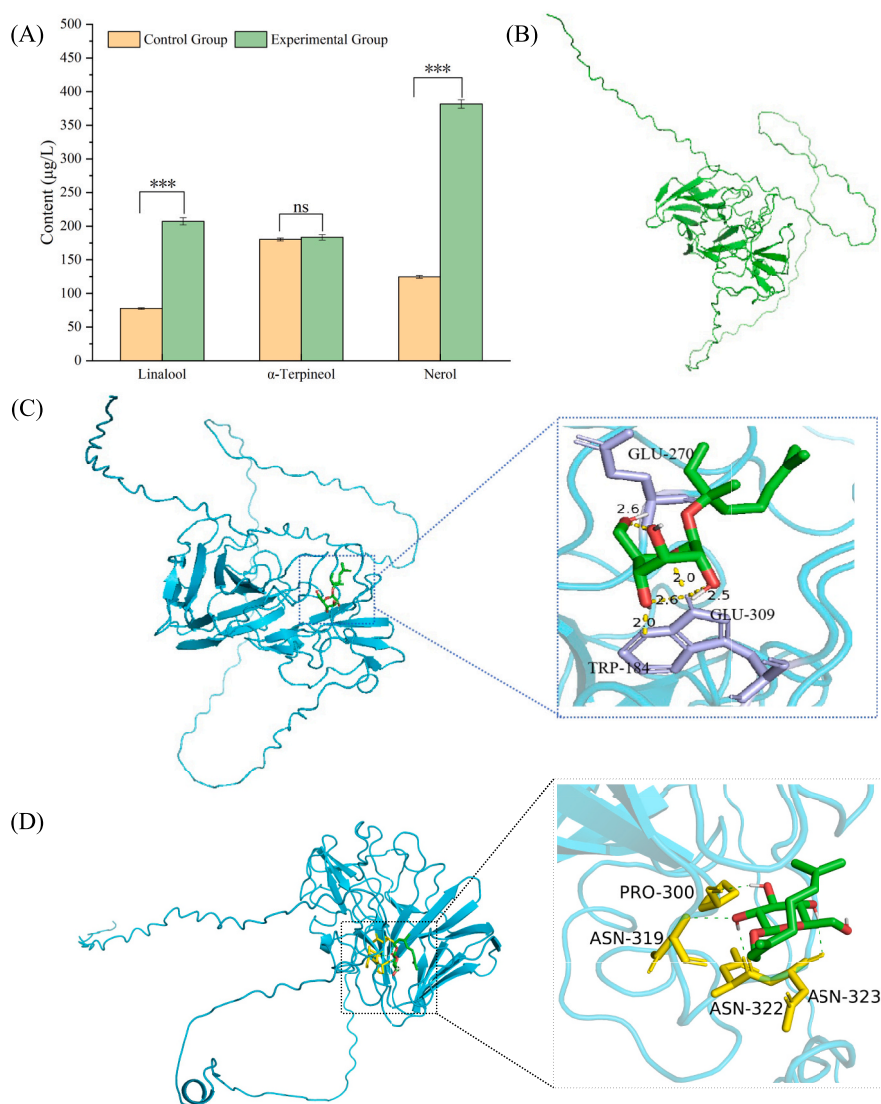


Fig. 5. Volatile terpene compounds detected in hop essential oil with and without *PkBGL* (A); Protein model of *PkBGL* (B); 3D binding pattern of *PkBGL* with linalyl glycoside (C) and nerol glycoside (D). “***” and “ns” indicate extremely highly significant ($p < 0.001$) and no significant ($p > 0.05$) differences, respectively.

results showed that their binding free energy was -6.0 and -6.2 kcal/mol, respectively. Linalyl glycoside formed hydrogen bonds with Glu270, Trp184, and Glu309 (Fig. 5C), while nerol glycoside interacted with Pro300, Asn319, Asn322, and Asn323 (Fig. 5D). Both glycosides were observed to bind to *PkBGL* through hydrogen bonding interactions and hydrophobic interactions with other amino acids around the hydrophobic pocket. This indicated that the complex formed by both the linalyl and nerol glycosides and *PkBGL* was stabilized not only by hydrophobic interaction forces but also by hydrogen bonding (Jiang et al., 2024). Therefore, we speculated that the high hydrolytic capacity of *PkBGL* for releasing terpenes may stem from its ability to form a stable complex by interacting with glycosides. A comprehensive approach involving site-directed mutagenesis, the identification of structural nuances, and the mapping of functional domains is warranted in future studies to fully elucidate the structural-functional rationale of *PkBGL*.

4. Conclusions

In this study, 12 non-*Saccharomyces* yeasts were compared for potential utilization in beer, and *P. kudriavzevii* KD4 was selected as an effective strain for enhancing beer aroma via BGL activity. The co-fermentation protocol of KD4/US05 was optimized, and the result was

KD4/US05 at an inoculation ratio of 10:1, and US05 was inoculated 36 h after KD4, with a main fermentation period of 11 d, and an inoculation concentration of 1.1×10^7 CFU/mL. Aroma analysis of the co-fermented beer revealed that the utilization of KD4 significantly enhanced its floral and fruity aromas and created a unique aroma profile via the significant increase of terpenes and ester contents in the co-fermentation. The BGL from KD4 (*PkBGL*) was heterologously expressed in *P. pastoris* and purified. A comprehensive analysis of the enzymatic characteristics and protein conformations of *PkBGL* from the GH1 family was conducted. The *PkBGL* showed good low-pH, low-temperature, ethanol, and iso- α -acid tolerance, and the optimum pH was 5.5, the optimum temperature was 50°C . In simulated hop environments, *PkBGL* exhibited strong release capacity for terpene compounds. Molecular docking results of terpene glycosides with *PkBGL* further revealed the crucial role of hydrogen bonding and hydrophobic interactions in structural stability. This molecular insight may account for the ability of *PkBGL* to specifically hydrolyze glycosides and release terpenes. These findings highlight the potential of non-*Saccharomyces* yeasts and the BGL they produce to improve beer aroma profiles, contributing to the enrichment of more floral and fruity notes and higher sensory quality.

While our findings provide a robust framework for improving beer aroma, further research is required to fully bridge the gap toward large-

scale industrial application. Future studies should focus on validating the performance of strain KD4 across a broader range of beer styles (e.g., Lager or Stout) and assessing its genetic stability and metabolic reproducibility across multiple fermentation cycles under commercial conditions. Additionally, more in-depth investigations into the metabolic synergy between non-*Saccharomyces* and *Saccharomyces* yeasts, specifically the redistribution of carbon flux and nutrient competition, will further elucidate the complex mechanisms underlying enhanced ester production. These efforts will provide more comprehensive theoretical and practical foundations for the application of KD4 in the craft brewing industry.

CRedit authorship contribution statement

Lin Yuan: Writing – review & editing, Writing – original draft, Visualization, Conceptualization. **Xiaoping Hou:** Resources, Funding acquisition. **Yang He:** Resources, Funding acquisition. **Xiujuan Wan:** Resources, Funding acquisition. **Jiaxin Han:** Visualization. **Wenqi Xu:** Writing – review & editing. **Shuo Yu:** Methodology, Formal analysis. **Shuang Bi:** Resources, Funding acquisition. **Lijuan Ma:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization. **Lei Xing:** Supervision, Resources, Funding acquisition.

Ethical statement

All sensory evaluations were conducted in accordance with ethical principles ensuring the voluntary participation, informed consent, and confidentiality of all assessors. Participants were fully informed of the study's objectives, procedures, and potential uses of data, and consent was obtained prior to participation. No sensitive personal data were collected, and all data were anonymised and used solely for research purposes. Participants retained the right to withdraw at any time. The protocol received approval from the Ethics Committee of Tianjin University of Science and Technology, Tianjin, China, ensuring adherence to ethical standards in human research.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foodchem.2026.149138>.

Data availability

Data will be made available on request.

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